

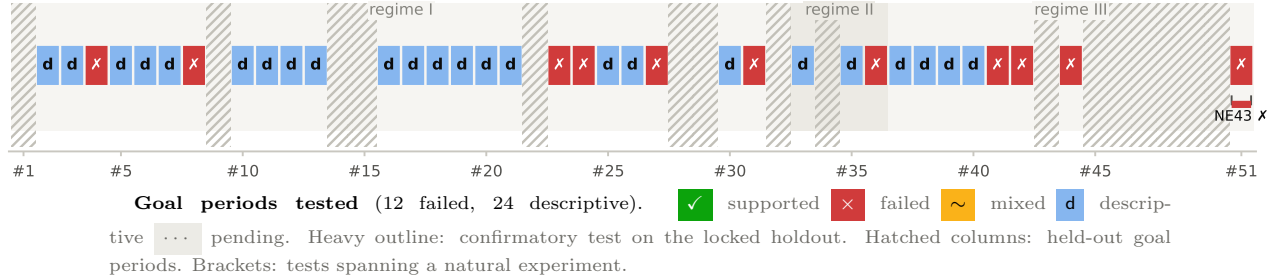
H43: Kicks leave a refractory window

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2.5/5

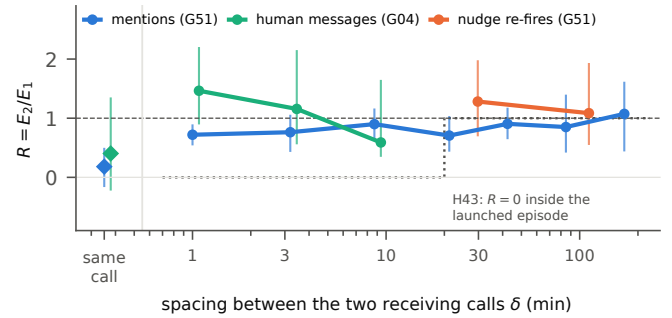
Bottom line. Kicks leave no refractory window: a second kick read by a later model call works about as well as the first; only kicks in the same read are wasted.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H43 asks whether a kicked agent ignores a second kick until the task that the first kick launched is over. Each kick is timed at the model call that first read it (DQ1 ledger). Past-only matching compares the second kick's effect, as a function of spacing δ , with an isolated first kick. The synthetic validation (planted windows) comes first. Replication covers 35 non-holdout periods, with native tests on NE43 (nudger off), G38 (pause gates) and G04 (human chat). The data show no refractory window. A second kick read by a later call keeps 80–130% of the effect at every δ from 1 min to 4 h. Mentions give $R = 0.78$ [0.67, 0.89] (15 periods), human messages $R \approx 1.4$, and nudge re-fires 1.28. Only a kick read in the *same* call adds little ($R = 0.2$ –0.4). The episode lock is absent (in-episode 0.67 vs 0.83). The holdout is not run.



Observables. Second-kick effect relative to a first kick (lnHR ratio, 95% CI), against the spacing of the two receiving calls. Diamonds: the second kick read in the *same* call. Dotted: H43's prediction ($R = 0$ inside a 20-min launched episode). Mentions (G51) dip only to ~ 0.8 (pooled over 15 periods: 0.78). Human messages (G04) and nudge re-fires (G51) stay at or above 1.

PHYSICAL MODEL

Model 09 (renewal or refractory Hawkes) with model 03's refractory compartment. A kick read at time t multiplies the agent's escape (or reply) hazard by a gain g_c with a short kernel κ . A recovery function r runs from the previous effective kick. H43 claims $r = 0$ inside the launched task episode of length L . The rivals are a habituation clock $r(\delta)$, $r \equiv 1$ (state dependence only), and read-out batching (kicks read in one call count once). The estimator $R(\delta) = E_2/E_1$ uses pooled log hazard ratios.

$$\lambda_{\text{esc}}(t) = \lambda_0(a) [1 + g_c \kappa(t - t_{\text{read}}) r(t - t_{\text{last}})]$$

$$\text{H43: } r = \mathbb{I}[t - t_{\text{last}} > L]; \quad \text{habituation: } r = 1 - a e^{-\delta/\tau}$$

$$E = \ln \frac{\sum \text{events}_K / \sum \text{at-risk}_K}{\sum \text{events}_C / \sum \text{at-risk}_C}$$

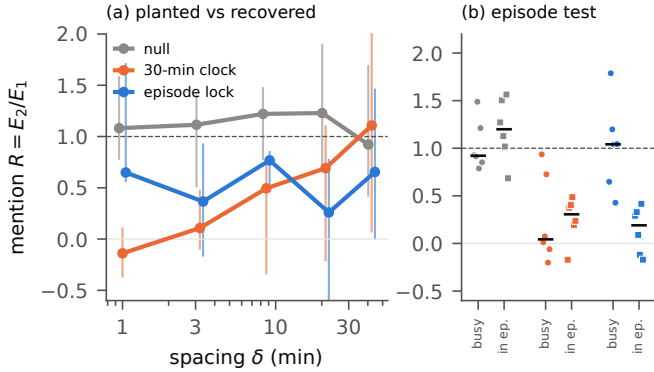
$$R(\delta) = E_2(\delta)/E_1, \quad \delta_{1/2} : R(\delta_{1/2}) = \frac{1}{2}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not wait for an agent to finish its task before you nudge or message it again. A kick read by a later call works about as well as the first (R 0.8–1.3). A kick sent before the agent's next model call, e.g. during its PAUSE, lands in the same read as the earlier one and adds little ($R = 0.2$ –0.4). Merge such messages, or deliver them after the next call. The nudger's 15-min floor is already past any refractoriness. Its real limit is that a first nudge buys a glance (escape $\times 1.75$), not sustained work. The physics label (refractory renewal) does not survive. The context assembly is the unit of attention.

RESULTS VS PREDICTIONS

Prediction	Observed	
P1 window, $R_{\text{short}} \leq 0.5$	mentions 0.57–0.79; humans, nudges ≥ 1	×
P2 window $\approx$ episode	$\delta_{1/2} = 0$ vs $\tilde{L} = 5\text{--}56$ min	×
P3 episode lock	R_{in} 0.65–0.69 vs R_{act} 0.81–0.84	×
P4 same-call kick ≤ 0.3	0.18 (reply), 0.31 (talk)	✓
P5 synthetic recovery	nulls clean; mention windows only	~
P6 NE43 nudge window	no sustained E_1 ; re-fire $R = 1.28$	×
P7 invariance across NE43	mention $\Delta R = \pm 0.16$	✓
P8 G38 pause gates	8 re-kicked gates: untestable	n/a
P9 G04 human trains	$R(\delta \leq 2 \text{ min}) = 1.46$	×
P10 rule reliable	no window in 98–100% of draws	✓



Second figure. Synthetic validation at G51 scale. (a) Mention $R(\delta)$: the null stays near 1. The estimator recovers a planted 30-min habituation clock ($\delta_{1/2}$ 15 vs 18 min); an episode lock gives an intermediate curve. (b) The episode test separates the three worlds. The real data look like the null in both panels.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 × failed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 × failed
G10 d descriptive
G11 d descriptive
G12 d descriptive

+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Only mention curves are identifiable at G51 size. Human messages are testable in G04 alone (5-min quiet rule, amended before outcomes), and nudges only marginally. The nudge first-kick effect exists for “any activity” (post hoc) but not for sustained work (pre-registered), so R for nudges rests on the post hoc outcome. Re-fired nudges may beat first ones because of the nudger’s unobserved text trigger. Estimated call starts for non-Gemini agents misclassify some same-call and next-call pairs; this pulls $R(\delta \leq 2)$ toward the batched value. Episodes come from minute-grid runs, not task content. With 35 periods $\times$ 3 classes $\times$ 5 outcomes, the replication verdicts are templated and not independent.

ROUND 2

Price the k -th kick inside one read (per-provider timing). Test nudge facilitation against the trigger (H35). Run content-defined episodes and a content-drift outcome. Run `confirm.py`.

- **Essence.** the unit of operator attention. Round 1 says it is the recipient’s read-out (one model call), not its task episode; round 2 should confirm that on the holdout and price the k -th kick inside one read.
- **H43-R1** Run the confirmatory script.
- **H43-R2** Nudge facilitation or nudger selection?
- **H43-R3** Read-out batching as the object.
- **H43-R4** Content-defined episodes and a content outcome.
- **H43-R5** Pooled gate test.
- **H43-R6** Decompose the NE43 swarm drop.